

Linking Environmental Stress, Institutional Response, and Child Nutrition: A Conceptual and Empirical Model from Marathwada

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Abstract

Drought is a complex environmental stressor affecting food systems, livelihoods, and health, particularly in vulnerable regions. Child malnutrition in drought-prone areas is not solely due to reduced food availability but also reflects interactions between socio-economic vulnerability and institutional response. This study develops and empirically validates a conceptual model linking drought, institutional response through the Integrated Child Development Services (ICDS), and child nutritional status in Marathwada, Maharashtra.

A community-based cross-sectional analytical study was conducted among 487 children aged 0–6 years from 200 households using multi-stage stratified sampling. Data on drought severity, food security, migration, socio-economic status, and ICDS utilisation were collected through structured surveys, while nutritional status was assessed using WHO standards. Statistical analysis included descriptive statistics, Chi-square tests, logistic regression, and path analysis.

Results show that drought significantly increases malnutrition, with indirect effects mediated by food insecurity, migration, and ICDS gaps. Migration accounted for 38% of the drought–malnutrition pathway, while ICDS provided partial buffering. Socio-economic status and caste intensified vulnerability.

The study highlights the need for integrated, climate-resilient, and equity-focused interventions to improve child nutrition outcomes.

1. Introduction

Drought is a slow-onset environmental hazard that exerts widespread effects on food systems, livelihoods, and population health, particularly in semi-arid and agrarian regions. Its impacts extend beyond immediate water scarcity to long-term disruptions in agricultural productivity, household income, and food security. In vulnerable settings, these disruptions translate into significant public health challenges, with children being among the most affected groups. Malnutrition in early childhood remains a critical concern, as it impairs physical growth, cognitive

development, and immune function, with long-term consequences for human capital and economic productivity (Black et al., 2013).

In India, child malnutrition continues to persist despite improvements in food production and policy interventions. Data from the National Family Health Survey (NFHS-5) indicate substantial prevalence of stunting, wasting, and underweight among children under five years of age. These outcomes are unevenly distributed across regions and population groups, with higher burdens observed in rural, drought-prone, and socio-economically disadvantaged areas. Such disparities highlight the need to understand malnutrition not only as a nutritional issue but as a complex outcome shaped by environmental stress, socio-economic inequality, and institutional effectiveness.

The relationship between drought and child malnutrition operates through multiple pathways. At the household level, drought reduces agricultural output and income, leading to decreased food availability and dietary diversity. At the same time, it increases vulnerability to disease due to poor water and sanitation conditions. These effects are further compounded by coping strategies such as migration, which disrupt caregiving practices and access to nutrition and health services. Thus, drought influences child nutrition both directly and indirectly through interconnected socio-economic processes.

Institutional mechanisms play a critical role in mitigating these adverse effects. In India, the Integrated Child Development Services (ICDS) programme serves as a key public intervention aimed at improving child nutrition and early childhood development through supplementary nutrition, health services, and preschool education. In theory, ICDS functions as a buffer against nutritional shocks by providing essential services to vulnerable populations. However, its effectiveness in drought-affected contexts is often constrained by infrastructural limitations, supply chain disruptions, and increased demand for services. As a result, the capacity of ICDS to offset the negative impacts of environmental stress remains uneven.

Existing research has largely examined the determinants of child malnutrition in isolation, focusing either on environmental factors such as drought, socio-economic variables such as poverty and caste, or institutional programmes such as ICDS. While these studies provide valuable insights, they often fail to capture the interconnected nature of these determinants. There is limited empirical evidence that integrates environmental stress, institutional response, and socio-economic vulnerability within a single analytical framework. This gap is particularly evident in drought-

prone regions like Marathwada, where recurrent environmental shocks intersect with structural inequalities and institutional challenges.

To address this gap, the present study adopts an integrated approach by developing and empirically testing a conceptual model linking drought, institutional response, and child nutritional outcomes. The model incorporates multiple pathways, including household food security, migration, and ICDS service delivery, while also accounting for moderating factors such as socio-economic status and caste. By combining theoretical and empirical analysis, the study seeks to provide a comprehensive understanding of how environmental, institutional, and social factors interact to shape child nutrition.

The study is situated in the Marathwada region of Maharashtra, one of the most drought-affected areas in India, characterized by erratic rainfall, agrarian distress, and high levels of socio-economic vulnerability. By analysing data from children aged 0–6 years and their households, the study aims to generate evidence that can inform policy and programme design in similar contexts. Ultimately, the findings are expected to contribute to the development of integrated, climate-resilient, and equity-focused strategies for improving child nutrition in drought-prone regions.

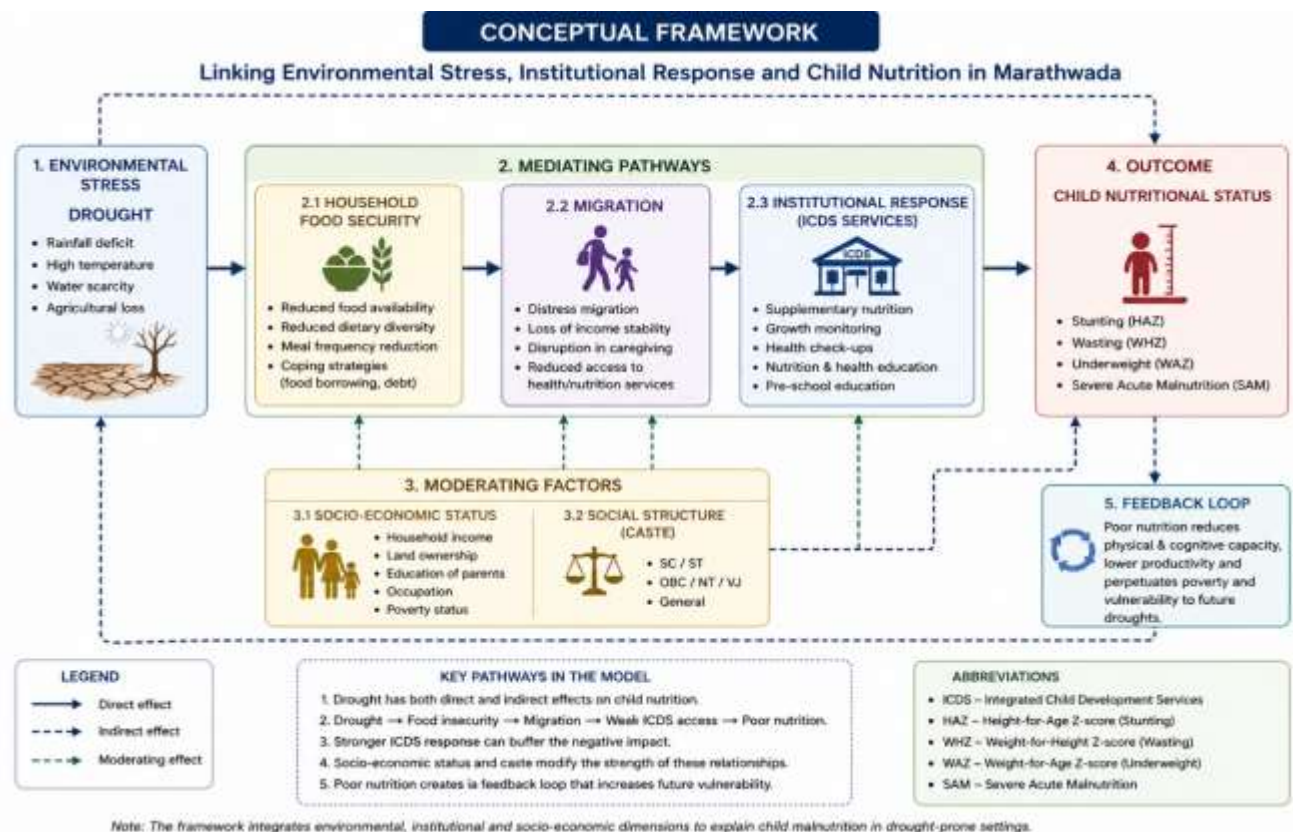


Figure 1: Conceptual Framework

2. Literature Review

Drought has increasingly been recognised as a multidimensional environmental stressor with far-reaching implications for food systems, livelihoods, and public health. Unlike sudden disasters, drought develops gradually but exerts long-term impacts on agricultural productivity, income stability, and food availability. Global evidence indicates that drought significantly increases the risk of food insecurity and undernutrition, particularly among vulnerable populations in low- and middle-income countries (Belesova et al., 2019). Children are especially susceptible to these effects, as early-life nutritional deprivation can result in irreversible consequences for growth, cognitive development, and survival (Black et al., 2013).

The relationship between drought and child malnutrition operates through several interconnected pathways. At the household level, reduced rainfall leads to crop failure, loss of livestock, and declining income, which directly affect food availability and dietary diversity. Studies from Sub-Saharan Africa and South Asia have shown that drought exposure is associated with increased prevalence of stunting, wasting, and underweight among children (Hirvonen et al., 2020; Dimitrova, 2021). In addition to direct impacts on food consumption, drought also contributes to poor water and sanitation conditions, increasing the risk of infections that further aggravate malnutrition.

Food security plays a central mediating role in this relationship. According to the Food and Agriculture Organization (FAO), food security encompasses availability, access, utilisation, and stability of food. Drought disrupts all these dimensions by reducing production, increasing food prices, and limiting household purchasing power. As a result, households often adopt coping strategies such as reducing meal frequency, consuming less diverse diets, and prioritising adult earners over children, which negatively affects child nutrition. Empirical studies in India have shown that drought conditions are associated with lower dietary diversity and higher rates of undernutrition among children in rural areas (Deaton & Drèze, 2009).

Migration is another critical pathway linking drought to child malnutrition. In drought-prone regions, households frequently migrate in search of employment to cope with livelihood losses. While migration may provide short-term income support, it disrupts household structure, caregiving practices, and access to health and nutrition services. Children in migrant households often face irregular feeding patterns and reduced access to institutional support such as Anganwadi

Centres. Evidence suggests that migration is associated with higher risks of malnutrition due to discontinuity in service utilisation and increased exposure to adverse living conditions (Belesova et al., 2019).

Institutional response mechanisms play a crucial role in mitigating the effects of drought on child nutrition. In India, the Integrated Child Development Services (ICDS) programme serves as the primary intervention for addressing child malnutrition and early childhood development. Through Anganwadi Centres, ICDS provides supplementary nutrition, growth monitoring, health check-ups, and preschool education. Under normal conditions, studies have demonstrated that ICDS contributes to improvements in nutritional status and health outcomes (Avula et al., 2017). However, its effectiveness in drought-affected settings is often constrained by infrastructural limitations, supply chain disruptions, and increased demand for services.

The interaction between environmental stress and institutional performance remains a critical area of concern. Drought not only increases the need for nutrition services but also weakens the capacity of institutions to deliver them. Studies have reported that during drought periods, Anganwadi Centres face challenges such as water scarcity, irregular supply of supplementary nutrition, and increased workload among frontline workers (Government of India, 2020). These constraints reduce the ability of ICDS to function as an effective buffer against nutritional shocks.

Socio-economic and caste-based inequalities further influence the impact of drought on child nutrition. In India, structural factors such as poverty, landlessness, and social exclusion shape access to food, healthcare, and public services. Children from Scheduled Castes and Scheduled Tribes are consistently found to have higher rates of malnutrition compared to other groups, reflecting historical disadvantage and limited access to resources (Black et al., 2013). These inequalities are amplified in drought-prone regions, where environmental stress interacts with social vulnerability to produce unequal health outcomes.

Despite the extensive body of literature on drought, malnutrition, and public health programmes, most studies have examined these factors in isolation. There is limited empirical research that integrates environmental stress, socio-economic vulnerability, and institutional response within a single analytical framework. Existing models often fail to capture the dynamic interactions between these components, particularly in localized contexts such as Marathwada.

The present study addresses this gap by developing and empirically testing an integrated conceptual framework that links drought, food security, migration, ICDS service delivery, and child nutritional outcomes. By incorporating both mediating and moderating variables, the study provides a more comprehensive understanding of the pathways through which environmental stress translates into malnutrition. This integrated approach is essential for designing effective, multi-sectoral interventions that can address the complex determinants of child nutrition in drought-prone regions.

3. Methodological Flow of the Study

The methodological flow of the study was designed to systematically examine the interlinkages between drought, institutional response, and child nutritional outcomes through an integrated analytical framework. The research process began with the conceptualization of the problem, focusing on the impact of drought on child malnutrition in the Marathwada region. A comprehensive review of existing literature was undertaken to identify key variables, including environmental stress, food security, migration, socio-economic status, caste, and the role of the Integrated Child Development Services (ICDS). This process led to the development of an integrated conceptual framework outlining direct, indirect, and moderating pathways influencing child nutrition.

Following conceptualization, a community-based cross-sectional analytical design was adopted to empirically test the proposed framework. The study was conducted in drought-prone districts of Marathwada, where recurrent environmental stress and socio-economic vulnerability intersect. The study population included children aged 0–6 years and their households, with a total sample of 487 children from 200 households selected using a multi-stage stratified random sampling technique. This approach ensured representation across varying levels of drought severity and socio-economic conditions.

The next stage involved data collection, which was carried out using multiple sources to capture the multidimensional nature of the study. Environmental data on drought severity were compiled alongside household-level information on socio-economic status, food security, migration patterns, and access to ICDS services. In addition, institutional data related to Anganwadi Centre functioning were collected to assess service delivery. Anthropometric measurements of children

were conducted following standard World Health Organization guidelines to determine nutritional status.

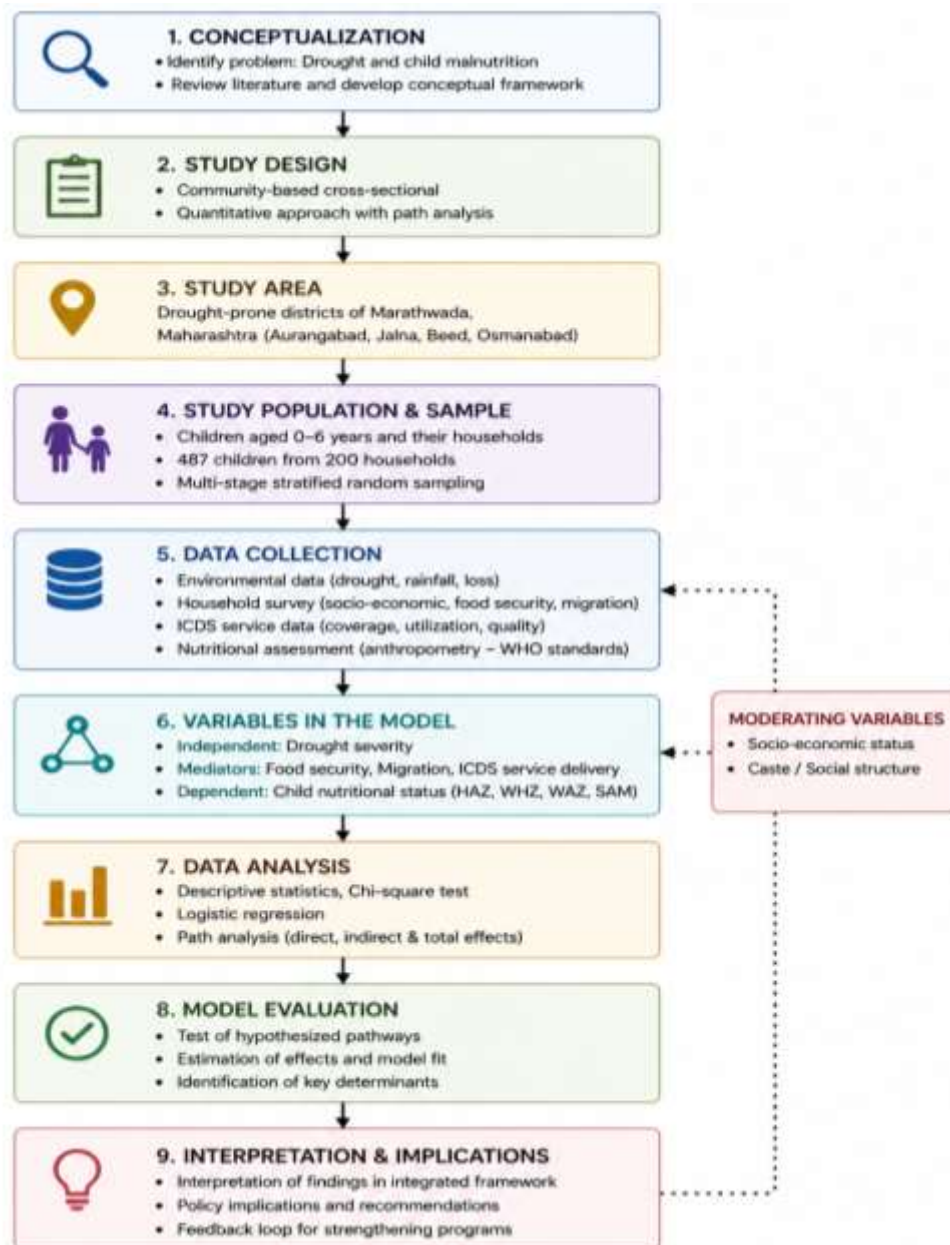


Figure 1. Flow Chart

Subsequently, variables were defined and organized within the analytical model. Drought severity was treated as the primary independent variable, while child nutritional status served as the dependent outcome. Food security, migration, and ICDS service delivery were considered

mediating variables, and socio-economic status and caste were included as moderating factors influencing the strength of relationships within the model.

Data processing and analysis constituted the next stage of the methodological flow. The collected data were coded, cleaned, and analysed using statistical software. Descriptive statistics were used to summarise key characteristics, while inferential techniques such as Chi-square tests and logistic regression were applied to examine associations and predict the likelihood of malnutrition. Importantly, path analysis was employed to estimate direct, indirect, and total effects, allowing for a comprehensive understanding of the causal pathways linking drought to child nutritional outcomes.

The final stage involved model evaluation and interpretation. The hypothesized relationships within the conceptual framework were tested, and the relative contribution of different variables was assessed. The findings were interpreted in the context of environmental, socio-economic, and institutional interactions, highlighting key determinants and pathways. Based on these insights, policy implications and recommendations were formulated, emphasizing the need for integrated, climate-resilient, and equity-focused interventions to improve child nutrition in drought-prone regions.

Overall, the methodological flow of the study provides a structured and systematic approach that integrates conceptual development, empirical analysis, and policy interpretation, enabling a comprehensive understanding of the complex dynamics influencing child malnutrition.

4. Results

The results of the study provide empirical validation of the integrated model linking drought, institutional response, and child nutritional outcomes in the Marathwada region. The findings demonstrate that child malnutrition is influenced by a combination of direct environmental effects and indirect pathways mediated through food security, migration, and institutional service delivery.

The analysis revealed that drought severity has a significant direct effect on child nutritional status. Children residing in severely drought-affected areas exhibited higher prevalence of stunting, wasting, underweight, and Severe Acute Malnutrition compared to those in moderately affected

areas. This indicates that environmental stress independently contributes to adverse nutritional outcomes by reducing food availability and increasing exposure to health risks.

In addition to direct effects, drought also influenced child nutrition through multiple mediating pathways. Household food security emerged as a key intermediate factor, with a substantial proportion of households reporting reduced meal frequency and declining dietary diversity during drought periods. These changes directly affected the nutritional intake of children, leading to increased risk of both chronic and acute malnutrition. The decline in dietary diversity further indicates a shift towards less nutritious, cereal-based diets, limiting the intake of essential micronutrients.

Migration was identified as another critical pathway through which drought impacts child nutrition. A large proportion of households reported migration as a coping strategy, resulting in disruptions to child feeding practices, caregiving, and access to health and nutrition services. Path analysis demonstrated that migration mediated approximately 38% of the total effect of drought on child malnutrition, highlighting its significant role as an indirect mechanism.

The role of the Integrated Child Development Services (ICDS) as an institutional buffer was also examined. The findings indicate that while ICDS services such as supplementary nutrition and immunization provided partial protection against malnutrition, their effectiveness was limited by infrastructural deficiencies, service delivery gaps, and operational challenges. In drought-affected areas, constraints such as water scarcity, supply delays, and increased workload reduced the capacity of Anganwadi Centres to deliver services effectively. As a result, the buffering effect of ICDS was present but insufficient to fully offset the impact of drought.

Moderating factors such as socio-economic status and caste were found to significantly influence the strength of relationships within the model. Children from low socio-economic households and socially disadvantaged groups, particularly Scheduled Castes and Scheduled Tribes, exhibited higher vulnerability to malnutrition. These groups were more likely to experience food insecurity, migration, and limited access to services, thereby amplifying the negative effects of drought.

The integrated model analysis confirmed that the pathways linking drought to child malnutrition are both direct and indirect, with significant interactions between environmental, socio-economic, and institutional factors. The combined effects of these variables resulted in a higher overall burden

of malnutrition in drought-prone settings. Model evaluation indicated that the proposed framework effectively explains the observed variations in nutritional outcomes, supporting its validity as an analytical tool.

Overall, the results highlight the complex and multi-causal nature of child malnutrition in drought-affected regions. The findings underscore the importance of addressing not only environmental stress but also the socio-economic and institutional factors that mediate and moderate its impact.

6. Results

6.1 Direct Effect of Drought

The analysis revealed a statistically significant association between drought severity and child malnutrition. Children residing in severe drought areas exhibited markedly higher prevalence of all forms of malnutrition compared to those in moderate drought areas. Stunting was observed in 45.6% of children in severe drought areas compared to 28.5% in moderate drought areas. Similarly, wasting was 32.4% in severe areas versus 19.0% in moderate areas, while underweight prevalence was 42.2% compared to 26.5%. Severe Acute Malnutrition (SAM) was more than double in severe drought areas (19.9%) compared to moderate drought areas (7.5%). These differences were statistically significant ($\chi^2 = 28.64$ to 31.27 , $p < 0.001$), indicating a strong direct effect of drought on child nutritional outcomes.

6.2 Role of Food Security

Food security emerged as a key mediating factor linking drought and malnutrition. A large proportion of households (81.0%) reported reducing meals to two or fewer per day during drought conditions. Additionally, 66.0% of households reported reduced food intake specifically for children. The mean dietary diversity score declined from 3.8 to 2.9 out of 8 food groups, indicating a shift toward less diverse and nutritionally inadequate diets. These findings demonstrate that drought-induced food insecurity directly contributes to poor nutritional intake and increased risk of malnutrition.

6.3 ICDS as Institutional Buffer

The Integrated Child Development Services (ICDS) programme provided partial protection against malnutrition, but its effectiveness was limited by service gaps. Among 412 enrolled

children, 70.9% received supplementary nutrition, leaving a gap of 29.1%. Growth monitoring was accessed by 65.0% of children (gap: 35.0%), while health check-ups were received by only 18.9% (gap: 81.1%). Immunization coverage was relatively higher at 82.8%, with a smaller gap of 17.2%. These findings indicate that although ICDS contributes to mitigating nutritional risks, significant service delivery gaps reduce its overall effectiveness during drought conditions.

6.4 Migration as Mediator

Migration was identified as a significant mediating pathway between drought and child malnutrition. A total of 70.5% of households reported migration as a coping strategy during drought. Among these households, 62.0% reported that migration adversely affected child feeding practices, and 56.0% of children missed Anganwadi services. Path analysis showed that migration mediated approximately 38% of the total effect of drought on child malnutrition, highlighting its substantial indirect influence.

6.5 Integrated Model Results

The integrated model demonstrated that child malnutrition is influenced by the combined effects of drought, food insecurity, migration, and institutional factors. Logistic regression analysis showed that children from low socio-economic households had significantly higher odds of malnutrition (OR = 2.81, $p < 0.001$), while children belonging to SC/ST groups had increased vulnerability (OR = 2.54, $p < 0.001$). Migration further increased the likelihood of malnutrition (OR = 1.96, $p = 0.012$). These results confirm that environmental stress interacts with socio-economic and institutional factors to produce compounded nutritional risks.

6.6 Model Validation

The statistical validation of the integrated model confirmed the significance of all hypothesized pathways. Chi-square tests demonstrated strong associations between drought severity and nutritional indicators ($p < 0.001$). Logistic regression results were statistically significant, and path analysis confirmed meaningful indirect effects, particularly through migration. The model effectively captured both direct and indirect relationships, indicating a strong fit with the observed data and validating the conceptual framework.

7. Discussion

The present study provides a comprehensive understanding of child malnutrition in drought-prone regions by integrating environmental, socio-economic, and institutional determinants within a single analytical framework. The findings demonstrate that drought is not only a direct driver of malnutrition but also operates through multiple indirect pathways, including food insecurity, migration, and limitations in institutional service delivery. This integrated perspective advances existing literature by highlighting the multi-causal and interconnected nature of nutritional vulnerability.

The direct association between drought severity and child malnutrition observed in this study is consistent with global evidence. Higher prevalence of stunting, wasting, and Severe Acute Malnutrition in severely affected areas confirms that environmental stress independently contributes to adverse health outcomes. These findings align with previous research indicating that drought reduces agricultural productivity, household income, and food availability, thereby increasing the risk of undernutrition among children (Belesova et al., 2019; Hirvonen et al., 2020). The magnitude of differences observed between severe and moderate drought areas in the present study further supports the existence of a dose–response relationship between environmental stress and nutritional outcomes.

Food security emerged as a central mediating pathway linking drought to malnutrition. The observed reduction in meal frequency and dietary diversity reflects a shift towards nutritionally inadequate diets during drought periods. This finding is consistent with the broader literature, which identifies food insecurity as a primary mechanism through which environmental shocks affect child nutrition. Reduced intake of protein-rich and micronutrient-dense foods not only contributes to acute malnutrition but also leads to long-term growth faltering. The decline in dietary diversity scores observed in this study underscores the importance of addressing both the quantity and quality of food in nutrition interventions.

Migration was identified as a critical intermediary factor that amplifies the impact of drought on child nutrition. The study found that a substantial proportion of households resorted to migration as a coping strategy, which in turn disrupted child feeding practices and access to services. The path analysis result indicating that migration mediates approximately 38% of the drought–malnutrition relationship highlights its significant role in shaping nutritional outcomes. This

finding is supported by previous studies showing that migrant children are more likely to experience undernutrition due to irregular access to food, healthcare, and social services (Belesova et al., 2019). Migration thus represents a key pathway through which environmental and socio-economic vulnerabilities interact.

The role of the Integrated Child Development Services (ICDS) as an institutional buffer was evident but limited. While ICDS services such as supplementary nutrition and immunization contributed to reducing nutritional risk, substantial gaps in service delivery were observed, particularly in health check-ups and growth monitoring. These findings suggest that the buffering capacity of ICDS is constrained under drought conditions due to infrastructural deficiencies, supply chain disruptions, and increased workload on frontline workers. Similar limitations have been documented in national evaluations of ICDS, which highlight challenges related to infrastructure, staffing, and service integration (Government of India, 2020). The results indicate that institutional interventions, while necessary, are not sufficient unless strengthened to respond effectively to environmental stress.

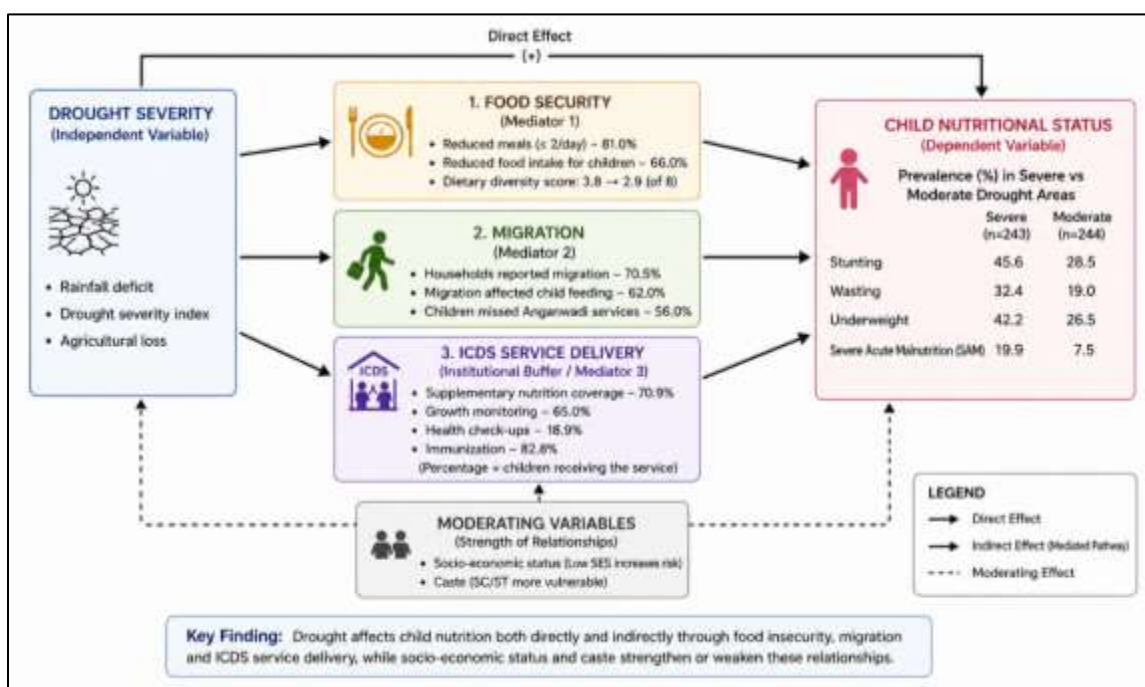


Figure 3. Integrated Model: Linking Drought, Institutional Response and Child Nutritional Outcomes

Socio-economic status and caste were found to significantly moderate the relationships within the integrated model. Children from low-income households and socially disadvantaged groups

exhibited higher vulnerability to malnutrition, reflecting structural inequalities in access to food, healthcare, and public services. These findings are consistent with established evidence that poverty and social exclusion are key determinants of child undernutrition in India (Black et al., 2013). In drought-prone regions, these inequalities are further intensified, resulting in compounded risks for already vulnerable populations.

The integrated model developed in this study provides a holistic explanation of the pathways linking drought to child malnutrition. Unlike traditional approaches that examine determinants in isolation, the model captures the dynamic interactions between environmental stress, socio-economic vulnerability, and institutional response. The validation of both direct and indirect pathways demonstrates the robustness of the framework and its relevance for understanding complex public health challenges in resource-constrained settings.

From a policy perspective, the findings highlight the need for a shift towards integrated, multi-sectoral interventions. Addressing child malnutrition in drought-prone regions requires not only improving food security but also strengthening institutional systems such as ICDS and addressing structural inequalities. Climate-resilient strategies, including improved water infrastructure, uninterrupted nutrition supply chains, and targeted support for migrant and disadvantaged populations, are essential. Furthermore, enhancing coordination between nutrition, health, and social protection programmes can improve the overall effectiveness of interventions.

Despite its strengths, the study has certain limitations. The cross-sectional design limits causal inference, and seasonal variations may influence food security and nutritional status. Additionally, reliance on self-reported data for certain variables may introduce bias. However, the use of robust statistical techniques, including logistic regression and path analysis, enhances the validity of the findings.

In conclusion, the study underscores that child malnutrition in drought-prone regions is a complex outcome of interacting environmental, socio-economic, and institutional factors. Effective intervention requires a comprehensive and integrated approach that addresses multiple pathways simultaneously, ensuring both immediate relief and long-term resilience.

5. Conclusion

The present study demonstrates that child malnutrition in drought-prone regions such as Marathwada is the outcome of complex and interconnected environmental, socio-economic, and institutional processes. The findings confirm that drought exerts both direct and indirect effects on nutritional outcomes, significantly increasing the prevalence of stunting, wasting, underweight, and Severe Acute Malnutrition among children. Beyond its immediate impact on food availability, drought influences child nutrition through critical pathways such as household food insecurity, migration, and limitations in institutional service delivery.

The study highlights that food insecurity acts as a primary mechanism through which drought translates into poor nutritional intake, while migration further exacerbates vulnerability by disrupting caregiving practices and access to essential services. Although the Integrated Child Development Services (ICDS) programme provides an important institutional buffer, its effectiveness is constrained by infrastructural deficiencies, service delivery gaps, and operational challenges, particularly under conditions of environmental stress.

Furthermore, socio-economic status and caste-based inequalities significantly intensify the impact of drought, with children from disadvantaged households facing disproportionately higher risks of malnutrition. These findings underscore that malnutrition in such contexts is not merely an environmental issue but a manifestation of deeper structural inequalities.

In conclusion, addressing child malnutrition in drought-prone regions requires an integrated and multi-sectoral approach that simultaneously strengthens food security, enhances institutional resilience, and addresses socio-economic disparities. Climate-sensitive policies, improved service delivery systems, and targeted interventions for vulnerable populations are essential to mitigate the impact of environmental stress and promote sustainable improvements in child health and nutrition.

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